THOMSON SIBUEA

Data Scientist | Full-Stack Developer | GeoAI & LLM Applications

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PROFESSIONAL SUMMARY

Data Scientist and IT Specialist at Statistics Indonesia (BPS) Lampung Province with a B.Sc. in Computational Statistics (GPA 3.70/4.00). Builds end-to-end data solutions, from data pipelines and machine learning models to the web platforms that deliver them. Experienced in machine learning, deep learning, remote sensing and GeoAI, and in LLM applications with Retrieval-Augmented Generation (RAG). Delivered 17 portfolio projects, including AI chatbots, recommendation systems, survey monitoring dashboards and a satellite-based forest-fire risk pipeline. Holds DataCamp Data Scientist Associate and AI Engineer for Data Scientists Associate certifications.

TECHNICAL SKILLS

Programming Languages: Python, R, SQL, JavaScript, TypeScript, PHP, HTML, CSS
Machine Learning & AI: Machine Learning, Deep Learning, NLP, LLM Applications, RAG, Sentiment Analysis, Recommendation Systems, TensorFlow, PyTorch, scikit-learn, OpenCV, LangChain, sentence-transformers, IndoBERT, Gemini API, Grok API
Data Analysis & Visualization: Pandas, NumPy, Jupyter, Matplotlib, Plotly, Streamlit, Tableau, Chart.js, Recharts, Looker Studio, Big Data, Statistics
Geospatial & Remote Sensing: Google Earth Engine, QGIS, GeoPandas, Rasterio, GDAL, PostGIS, Leaflet, Folium, Google Earth
Web Development: Next.js, React, Vue.js, Inertia.js, Flask, FastAPI, Django, Laravel, CodeIgniter 4, Node.js, Tailwind CSS, Bootstrap, Socket.IO, Prisma
Databases: MySQL, PostgreSQL, SQLite, Supabase, ChromaDB, Pinecone, Google Sheets
Tools & DevOps: Docker, Git, GitHub, Nginx, Postman, Google Cloud, Google Apps Script, Vercel, Microsoft Office

PROFESSIONAL EXPERIENCE

Junior Data Scientist / IT Specialist (IPDS Division) | 12/2024 - Present

Badan Pusat Statistik (BPS) Lampung Province, Indonesia
Government agency responsible for official statistics in Indonesia

- Build and maintain internal web applications and data platforms (SIMONA, TALITA, WAWAI, EksIm, queue and guest-book systems) used across BPS Lampung offices.
- Apply remote sensing, machine learning, deep learning and GeoAI to produce faster and more granular official statistics.
- Develop RAG-based assistants and LLM-powered analysis tools for statistics services and foreign-trade reporting.
- Deliver statistical processing and dissemination of 20+ reports, including 20+ high-quality statistical reports monthly, with accuracy and on-time release.
- Provide IT support to 15 internal agencies, maintaining 99.9% uptime for critical technology services.
- Created 5+ processing websites to streamline statistical and administrative tasks across the agency.
- Create and publish 30+ weekly data visualizations and 5-6 statistical infographics monthly to improve statistical literacy and public engagement.
- Directed protocol logistics for 5+ large-scale events related to the 2026 Economic Census and other sectoral statistical activities.

Information Technology Specialist (Internship) | 10/2024 - 11/2024

Badan Pusat Statistik (BPS) Head Office, Jakarta, Indonesia

- Contributed to migrating 500+ GB of national census and survey data from legacy servers to new server infrastructure.
- Migrated and verified Agricultural Census (ST), Economic Census (SE) and other national survey data to ensure completeness and consistency.
- Supported data integrity across nationwide datasets in line with government data security standards.

Data Analyst (Contract) | 03/2024 - 12/2024

Ministry of Industry of the Republic of Indonesia, Jakarta (Remote)

- Analyzed export and import trends for 50+ Harmonized System (HS) Codes and identified potential industries for Indonesia's future growth.
- Turned complex datasets into tables, visual reports and dashboards that supported data-driven decisions.
- Engaged 100+ companies to report monthly production activities, supporting institutional data compliance.

IT Technician (Internship) | 08/2023 - 10/2023

Badan Pusat Statistik (BPS) Cilegon City, Banten, Indonesia

- Entered and validated survey data from 50+ accommodation establishments, ensuring alignment with national statistical standards.

- Designed an honorarium management system for 30+ external partners using Google Form, Google Sheets and Looker Studio.
- Conducted field verification (ground checks) of Socio-Economic Registration (Regsosek) data across 5 sub-districts.
- Produced and edited a video for the Exemplary Statistician Award, highlighting staff achievements.

Web Developer | 12/2022 - 08/2023
Politeknik Statistika STIS, Jakarta, Indonesia
- Built a centralized web dashboard with CodeIgniter 4 to present results of four field practice (PKL) studies, improving data accessibility and reporting.

Data Analyst | 03/2023 - 06/2023
Politeknik Statistika STIS, Jakarta, Indonesia
- Developed 20+ graphs and tables from field practice research and integrated them into the dashboard for real-time access.

Field Data Verifier (Contract) | 02/2023
Badan Pusat Statistik (BPS), Jakarta, Indonesia
- Cross-verified and validated Socio-Economic Registration (Regsosek) data to maintain integrity and consistency.
- Conducted in-depth respondent interviews to authenticate survey answers and resolve discrepancies.
- Audited statistical outputs to confirm that reported data matched field conditions.

SELECTED PROJECTS

- **TALITA (StatistiKita), 2026:** AI-powered statistics service portal for BPS Lampung combining a RAG chatbot, consultation booking, live chat and a Q&A forum.
- **EksIm - Foreign Trade Dashboard, 2026:** Foreign-trade statistics dashboard automating BRS tables, with AI agents for analysis, policy and reports (Python, Streamlit, Grok API).
- **CERDAS - Candidate Recommendation System, 2025:** Semantic recommendation system matching civil-servant (CASN) candidates to positions by meaning (sentence-transformers, Pinecone).
- **SEHAT - Hospital & Doctor Recommendation, 2025:** Aspect-based sentiment analysis on patient reviews with an AI chatbot (Python, NLP, IndoBERT).
- **Forest Fire Early Detection, Riau, 2026:** Satellite data pipeline with machine learning that classifies forest-fire risk and estimates burned area (Google Earth Engine).
- **SIMONA BPS, 2026:** Monitoring system turning survey, news and business-classification data into interactive dashboards.
- **WAWAI, 2026:** Employee profile, CV and HR platform for all BPS Lampung offices, integrated with SIMPEG, with ATS-friendly CV export.
- **ESG Evidence Extraction, 2026:** Extracts ESG evidence from banks' sustainability reports against a scoring checklist using hybrid retrieval and RAG.
- More projects at thomsonsibuea.vercel.app/projects

EDUCATION

Bachelor of Applied Science in Computational Statistics (S.Tr.Stat.), GPA 3.70 / 4.00
Politeknik Statistika STIS, Jakarta, Indonesia | 09/2020 - 08/2024
- Thesis: Estimation of Potential Drought Areas Using Machine Learning Approach: Satellite Imagery.
- Activities: Komputasi.Net (Programming), Descent STIS (Data Science).

Bachelor's Degree in Ocean Engineering (first year), GPA 3.58 / 4.00
Institut Teknologi Sepuluh Nopember (ITS), Surabaya, Indonesia | 09/2019 - 06/2020

CERTIFICATIONS

- DataCamp: AI Engineer for Data Scientists Associate (11/2026, valid to 11/2028)
- DataCamp: Data Scientist Associate (11/2026, valid to 11/2028)
- Anthropic Claude Academy: AI Fluency: Framework and Foundations (11/2026)
- Anthropic Claude Academy: AI Fluency for Builders (11/2026)
- freeCodeCamp: Data Analysis with Python (07/2024)
- Kaggle: Intro to Deep Learning (06/2024), Computer Vision (06/2024), Intro to Machine Learning (04/2023), Data Visualization (03/2023)

LANGUAGES & ADDITIONAL SKILLS

Languages: Indonesian (Native), English (Proficient)
Additional Skills: Analytical Problem-Solving, Stakeholder Management, Strategic Communication, Data Integrity and Quality Assurance, Project Coordination, Adaptability